



Lab Experiment 06:

Simulation of Shipping Information using IoT

Problem Statement

It is required to realize shipping port products launching. Once a truck arrives at the departure gate, the shipping information is published to the MQTT broker. The information is obtained through barcode or automatically sent. The message consists of 3 fields. The message consists of (Serial Number, Port, Year, Month, Day, Hour, Minute, Destination, Type, Weight). The control center gathers all messages from all publishers by subscribing to the topics of different publishers. The broker server host the broker software (PicoMQTT).

Details

Publisher

Two publisher groups, one for Alex port and one for Portsaid port using the same publisher code with the only difference the topic indicates the port (eg. “port_alex_info” and “port_portasaid_info”). Each publisher has the option to choose whether get the truck info using barcode or automatically from a separate device (Arduino Uno). The publisher has to display the published message on the screen and log it in a log file.

Subscriber

Upon receiving a message has to:

1. Display it on the screen.
2. Speak out the sending port, destination, cargo weight using text to speech code provided.
3. Log the message to the log file.

Used Software

1. [Broker](#)
2. 2 editions of publishers one for [automatic truck info](#) and one for [barcode manual reading](#).
3. Arduino code for automatic [truck info](#).
4. [Subscriber](#)

Hardware

1. 3 notebooks (2 publishers + 1 subscriber)
2. 2 microcontrollers (broker server and automatic truck info using Arduino Uno).

Notes

1. Barcode using notebook (to be discussed in session).
2. Text to speech (to be discussed in session).

Delivery Policy

- Each group must send a 20-second video showing over all scenario for sending the details.
- Each group must send a log file for half an hour
- You should submit a report showing your schematic diagram and the challenges you faced (if any).
- You should cite any additional resources you used.
- Further details for the submission instructions will be posted later on MS Teams.

Good Luck